

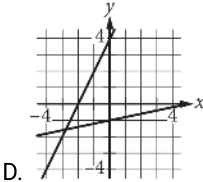
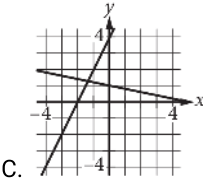
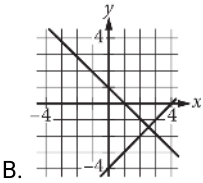
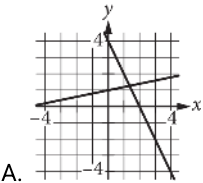
Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$x + 5y = 5$$
$$2x - y = -4$$

Which of the following graphs in the xy-plane could be used to solve the system of equations above?

Answer



Correct Answer: C

Rationale

Choice C is correct. The graph of a system of equations is the graph that shows the lines represented by each of the equations in the system. The x-intercept of the graph of each given equation can be found by substituting 0 for y in each equation: $x + 5(0) = 5$, or $x = 5$, and $2x - 0 = -4$, or $x = -2$. The y-intercept of the graph of each equation can be found by substituting 0 for x in each equation: $0 + 5y = 5$, or $y = 1$, and $2(0) - y = -4$ or $y = 4$. Using these x- and y- intercept values, the line that has equation $x + 5y = 5$ passes through the points (0,1) and (5,0), and the line that has equation $2x - y = -4$ passes through the points (0,4) and (-2,0). Only the lines in choice C pass through these points and can be used to solve the given system of equations.

Choices A, B, and D are incorrect. In choices A and B, neither line passes through (0,1) and (5,0) or (0,4) and (-2,0). In choice D, although one line passes through (0,4) and (-2,0) the other line doesn't pass through (0,1) and (5,0).

